**Exercise 1 (Semantics of CTL modalities).**

Let $\rho = s_0 s_1 s_2 \ldots$. Prove the following:

1. $\rho \models \Diamond\Phi$ iff there exists some $j \geq 0$ such that $s_j \models \Phi$
2. $s \models \exists\Box\Phi$ iff there is a path $\rho \in \mathrm{Path}(s)$ such that $\rho[j] \models \Phi$ for all $j \geq 0$
3. $s \models \forall\Box\Phi$ iff for all paths $\rho \in \mathrm{Path}(s)$ such that $\rho[j] \models \Phi$ for all $j \geq 0$

1) $\rho \models \Diamond \Phi$ iff $\rho \models \top \cup \Phi$ (def of $\Diamond$)

(def of $\cup$)

iff there exists $j \geq 0$ with $\rho[j\ldots] \models \Phi$ and for all $0 \leq i < j$: $\underbrace{\sigma[i\ldots] \models \top}_{\text{always true}}$

iff there exists $j \geq 0$ with $\rho[j\ldots] \models \Phi$

2) $s \models \exists \Box \Phi$ iff for some path $\rho \in Path(s) \models \Box \Phi$ Def of $\exists$

$\rho \models \Box \Phi$ iff for all $j \geq 0$, $\rho[j] \models \Phi$ Def of $\Box$

$\Rightarrow$ $s \models \exists \Box \Phi$ iff there is a path $\rho \in Path(s)$ such that $\rho[j] \models \Phi$ for all $j \geq 0$.

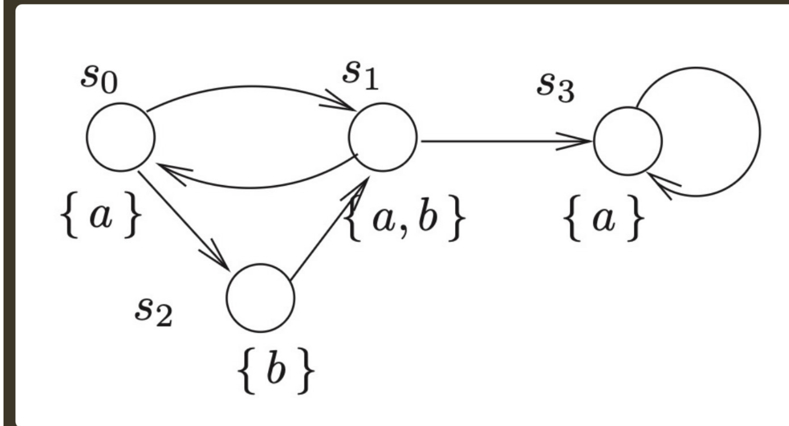
3) $s \models \forall \Box \Phi$ iff for all paths $\rho \in Path(s) \models \Box \Phi$ Def of $\forall$

$\rho \models \Box \Phi$ iff for all $j \geq 0$, $\rho[j] \models \Phi$

$\Rightarrow$ $s \models \forall \Box \Phi$ iff for all paths $\rho \in Path(s)$ such that $\rho[j] \models \Phi$ for all $j \geq 0$.

**Exercise 2 (CTL properties of transition systems).**

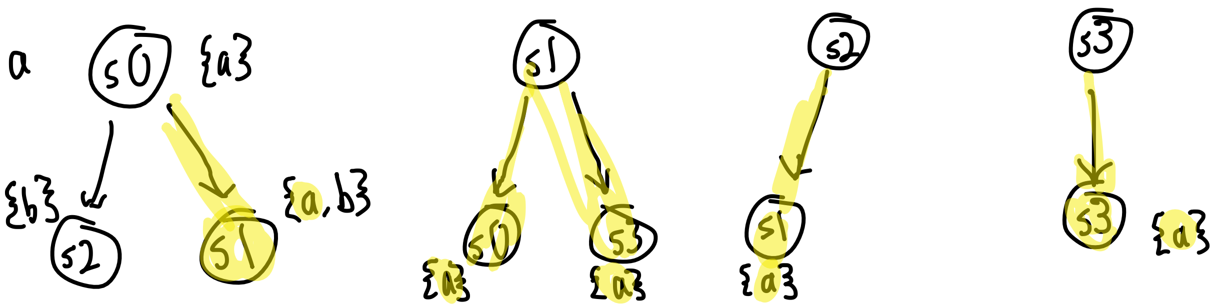
Consider the following transition system $\mathcal{T} = (S, \Sigma, \rightarrow, I, A, \lambda)$, with $I = \{s_0\}$:




For each of the following properties, decide for which states they hold. Does $\mathcal{T}$ satisfy them? Give arguments or counterarguments:

1. $\exists \bigcirc a$
2. $\forall \Box a$
3. $\exists \Diamond (\exists \Box a)$

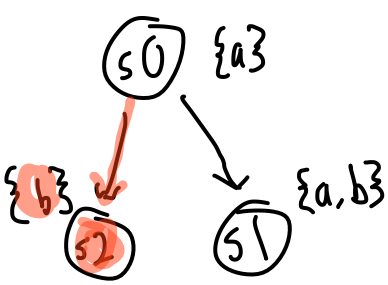
1) $\exists \bigcirc a$




All states satisfy $\exists \bigcirc a$ (as depicted by highlighted paths).

Since the initial states (s0) $\models \exists \bigcirc a$, $\mathcal{T} \models \exists \bigcirc a$

2) $\forall \Box a$




As we can see there exists a path where $\rho \models \Box a$ $\Rightarrow$ s0 $\not\models \forall \Box a$. This also means states capable of reaching s0 $\not\models \forall \Box a$ either $\Rightarrow$ s0, s1, s2 $\not\models \Box a$. Alternatively, s3 cannot reach any of those states ($\rho$ = s3s3s3...) and satisfies a $\Rightarrow$ s3 $\models \forall \Box a$. s0 $\not\models \forall \Box a \Rightarrow \mathcal{T} \not\models \forall \Box a$

3) $\exists \Diamond (\exists \Box a)$

Basically "there is some path we can take to reach a state where there is some path where $\Box a$ holds"

All states satisfy this property since for all states there exists a path to s3 which statisfies $\Box a$. $\tau \models \exists \Diamond (\exists \Box a)$ since $s0 \models \exists \Diamond (\exists \Box a)$